Dynamic Multi-Layer Asset Topology and Cluster Migration Under Geopolitical Stress

A Regime-Aware Framework for Detecting Structural Breaks in Cross-Asset Dependency Networks

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Abstract

This paper presents a dynamic multi-layer network framework we developed for tracking how global asset relationships reorganize during periods of geopolitical stress. The motivation came from a straightforward observation: traditional correlation-based analysis consistently fails to capture what is actually happening in markets when a crisis unfolds. Rather than relying on a single correlation matrix, our approach constructs three parallel dependency layers (linear shrinkage correlation, nonlinear distance correlation, and empirical tail co-exceedance), applies Leiden community detection over rolling windows, and introduces two novel quantitative metrics for measuring structural change. The first is a permutation-invariant Cluster Migration Index (CMI) built on Hungarian matching, and the second is a z-score normalized Topology Deformation Score (TDS) that fuses degree distribution shifts, community divergence, and spectral distance into a single composite. On top of this, we implemented transfer entropy and Granger causality to determine directional information flow between assets across different market regimes. Running the full pipeline on a 96-ETF universe covering equities, fixed income, commodities, currencies, crypto, managed futures, and 18 country-specific

funds from January 2010 through March 2026, one pattern kept appearing across every event window we studied: markets begin restructuring *before* the geopolitical event, not during it. Tail dependence picks up the signal first. Correlation catches up later.

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1 Introduction and Motivation

Most cross-asset analysis boils down to some variation of a correlation matrix. Compute pairwise Pearson correlations, maybe visualize it as a heatmap, and that is more or less the standard playbook. The issue with this is pretty fundamental: correlation is a linear measure, and markets do not break down in linear ways.

Take COVID as an example. Correlations across the board shot up to nearly 1.0 for a brief stretch, then reverted almost as quickly. Everything moved together, then stopped moving together. On the surface that looks like a temporary shock with no lasting structural change. But when we actually examined the community structure of the asset network through that period, we found something different: every single asset changed its cluster assignment. The entire topology dissolved and reformed. A Pearson heatmap does not tell you that story.

Then look at the Iran-Israel escalation cycle in 2024-2025. Pearson correlation barely flinched during some of these events. If that was your only lens, you would have concluded nothing structurally interesting happened. But the nonlinear and tail-dependence layers told a completely different story, with massive restructuring happening underneath the surface that linear correlation was blind to.

These observations are what pushed us to build this framework. We had a few questions we wanted to answer:

1. Do asset clusters genuinely restructure during geopolitical events, or does everything just co-move temporarily and snap back?

2. Is it possible to detect that restructuring *before* the event actually hits, using dependency measures beyond standard correlation?

3. Does structural change propagate in a specific causal direction across different dependency layers?

4. Which assets sit at the boundaries between clusters and serve as contagion pathways

during stress?

There is a decent body of literature on financial networks and community detection, but the way most of it treats cluster migration is as an afterthought of rolling correlation analysis. Nobody was really treating it as a standalone, measurable signal with its own dynamics and predictive content. And the standard method of comparing clusterings across time has a well-known but rarely addressed flaw: community detection algorithms like Leiden assign labels arbitrarily. Run the same algorithm on two identical structures and you might get labels $\{0, 1, 2\}$ one time and $\{2, 0, 1\}$ the next. A naive comparison interprets that as every asset having migrated, when in reality nothing changed at all.

That label permutation problem alone was introducing so much noise into the existing literature's migration measurements that we felt it warranted a ground-up rethinking of how these metrics should be computed. So we did that, built the full pipeline from data ingestion through topology export, and ran it across 16 years of daily data spanning 8 major geopolitical events.

2 Data and Universe

2.1 Asset Universe

Our universe consists of 96 ETFs, selected specifically for cross-asset breadth. The whole point of studying cluster migration is to see assets move between structurally different groups, so if you populate your universe with 96 flavors of US large-cap equity, you will not learn much. We needed genuine diversity: Treasuries that rally on flight-to-safety flows, commodities that respond to supply disruptions, EM currencies that get crushed during dollar strength, and so on.

Table 1: Asset Universe Composition (96 ETFs)

Category	**Count**	**Examples**
US Equity & Value	9	SPY, QQQ, IWM, DIA, VTI, SCHD
US Sectors	13	XLE, XLF, XLV, XLK, XLU, SOXX
International Equity	8	EFA, EEM, FXI, EWZ, EWJ
Country ETFs	18	EIS, INDA, EWG, EWT, ARGT, COLO
Fixed Income	8	TLT, IEF, SHY, LQD, HYG, EMB
Commodities	10	GLD, SLV, USO, DBA, URA, COPX
FX & Volatility	4	UUP, FXE, FXY, VIXY
Thematic & Defense	4	ITA, XAR, QTUM, BLOK
Global X Thematic	16	BOTZ, LIT, AIQ, BUG, CLOU
Managed Futures	4	DBMF, KMLM, CTA, WTMF
Crypto	2	BITO, IBIT

The 18 country-specific ETFs were especially critical for our geopolitical event studies. Having EIS (Israel) in the universe lets us directly observe how the Israeli equity market repositions relative to global clusters during the Iran-Israel conflict cycle. Same logic for INDA, EWG, EWT, and the others.

All data comes from the Financial Modeling Prep (FMP) API on their Ultimate tier plan. FMP is a financial data provider that offers institutional-grade market data through a REST API, covering equities, ETFs, fixed income, economic indicators, and more. We use their stable endpoint (`historical-price-eod/dividend-adjusted`) which returns dividend-adjusted daily closing prices going back to each ETF's inception date. The Ultimate plan gives us up to 3,000 API calls per minute and 150 GB of bandwidth, which is more than enough to fetch and maintain the full 96-ticker universe with room to scale. Our client implements async HTTP requests with a token-bucket rate limiter, parquet-based caching so we only re-fetch stale data (anything older than one business day), and automatic retry with exponential backoff.

2.2 Data Processing

The raw data goes through several cleaning steps before anything else touches it. We compute log returns and apply 1% Winsorization to handle extreme outliers without removing them entirely. Forward-fill handles gaps up to 5 trading days (holidays, exchange closures), and any ETF with more than 5% of its data missing gets automatically excluded from that particular rolling window.

Out of the 96 ETFs in the universe, 59 survived the full cleaning process across the entire 16-year sample. The other 37 are newer funds (things like IBIT, which launched in January 2024, or SHLD from March 2024) that simply do not have enough history for the earlier rolling windows. They get included automatically once they accumulate sufficient data.

The final dataset covers 3,938 trading days from January 2010 through March 2026, producing 764 rolling windows at a 120-day width with a 5-day step.

3 Methodology

The framework is organized in layers, each feeding into the next. Some of the design choices here are straightforward; others came out of trial and error where our initial approach turned out to be wrong and we had to go back and fix things.

3.1 Layer 1: Similarity and Dependency Measures

This is the foundation, and it is where our approach diverges most from the standard playbook. For each rolling window, instead of computing a single correlation matrix, we compute three separate dependency measures in parallel:

Table 2: Dependency Layers

Layer	Method	What It Captures
Linear	Ledoit-Wolf shrinkage correlation	Standard co-movement
Nonlinear	Distance correlation (dCor)	Nonlinear dependence; zero iff independent
Tail	Empirical co-exceedance	Joint extreme moves (crisis co-movement)

Why shrinkage instead of raw Pearson? With 59 assets in a 120-day window, the sample correlation matrix is seriously noisy. The number of parameters you are estimating ($59 \times 58/2 = 1{,}711$ unique correlations) is not that far from the number of observations (120). Ledoit-Wolf shrinkage regularizes the estimate by pulling eigenvalues toward a structured target, producing a better-conditioned matrix. When you are building graphs from this matrix, conditioning matters.

Distance correlation, introduced by Székely et al. [1], fills a gap that Pearson cannot. It equals zero if and only if two variables are statistically independent, meaning it will catch nonlinear dependencies that Pearson is completely blind to. This turned out to be one of our most important design decisions. During the June 2025 Twelve-Day War, the Pearson layer showed virtually no structural change, but dCor picked up massive restructuring.

The tail-dependence layer uses empirical co-exceedance at configurable quantiles (we tested 1%, 3%, 5%, and 10%). This measures how frequently two assets have simultaneous extreme moves, which is exactly the kind of dependency that matters during a crisis but gets averaged away in normal-regime correlation estimates.

For the directional flow analysis (described in Layer 6), we also compute transfer entropy via the KSG k-nearest-neighbor estimator [9] and Granger causality with Bonferroni correction across lags plus ADF stationarity pre-checks.

3.2 Layer 2: Graph Construction

Each dependency matrix becomes a graph. Our primary construction method uses threshold graphs with top-K edges per node, where we tested $K \in \{3, 5, 7, 10\}$ as part of our

sensitivity analysis. We also construct MSTs (minimum spanning trees) for the structural backbone following Mantegna [11], and PMFGs (planar maximally filtered graphs) following Tumminello et al. [10] for richer filtered topology.

With three dependency layers, each producing its own graph per window, this gives us a multi-layer adjacency tensor $\mathbf{A}[\text{layer}, i, j]$ that can be analyzed both per-layer and jointly.

One bug we caught early: the MST construction was summing the upper and lower triangle entries of the symmetric similarity matrix, which doubles every edge weight. Sounds minor, but it was distorting the tree structure. The fix was to take the max of symmetric entries instead.

3.3 Layer 3: Community Detection

The primary community detection method is the Leiden algorithm [3], which is the successor to Louvain. Leiden guarantees connected communities (Louvain does not) and runs faster on large graphs. We sweep the resolution parameter from 0.3 to 2.0 to verify that our structural findings are not artifacts of one particular resolution choice.

Beyond single-layer Leiden, we implemented three additional clustering approaches for comparison and validation:

- **K-Means baseline** on the same rolling windows and data, giving us a standard ML baseline to benchmark against
- **Consensus clustering** aggregating 100 independent Leiden runs per window into a consensus matrix
- **Multiplex consensus** that looks for agreement across all three dependency layers simultaneously, with NMI tracking how well the layers agree at each timestep

That last one turned out to be quite useful. When all three layers produce similar community structures, the topology is stable and well-defined. When they disagree, the

network is in a structurally ambiguous state, and we found that this disagreement often precedes a migration event.

3.4 Layer 4: Regime Detection

To contextualize migration events, we need to know what "regime" the market is in. We fit a 3-state Gaussian Hidden Markov Model on a feature vector consisting of realized volatility, mean pairwise correlation, and cross-sectional return dispersion, all computed from each rolling window. The three states auto-order by volatility into calm, transition, and stress.

Across our full sample, the HMM allocates 89.7% of windows to the calm state, 5.1% to transition, and 5.2% to stress. As of our last data point (March 20, 2026), the model classifies the current regime as stress.

We supplement the HMM with PELT change-point detection for identifying discrete structural breaks in the metric time series. The original implementation used an ad-hoc penalty of $\log(n) \cdot \mathrm{Var}(x)$, which was flagging way too many changepoints. We corrected this to the proper BIC formulation: $d \cdot \log(n)$ where $d = 2$ accounts for mean and variance parameters per segment.

For validation, we run out-of-sample regime prediction using TimeSeriesSplit with a constrained Random Forest, making sure topology metrics can actually predict regime labels without any look-ahead contamination.

3.5 Layer 5: Novel Migration Metrics

This is the methodological core of the paper. We developed four metrics for quantifying cluster migration over time. Each one addresses a specific gap in how the existing literature handles temporal cluster analysis, and several required non-trivial methodological corrections during development.

3.5.1 CMI: Cluster Migration Index

The Cluster Migration Index measures what fraction of assets genuinely changed cluster membership between consecutive windows:

$$\mathrm{CMI}(t) = \frac{1}{N} \sum_{i=1}^{N} \left[1 - \delta(\sigma(c_i(t))\,,\ c_i(t-1))\right] \tag{1}$$

where σ is the Hungarian-optimal bijection between the current and previous label sets, obtained via `scipy.optimize.linear_sum_assignment` [12]. CMI is bounded $[0, 1]$: zero means no assets changed clusters, one means total dissolution.

The Hungarian matching is the critical piece here. Without it, the metric is essentially broken. Consider a simple case: Leiden produces clusters $\{0, 1, 2\}$ at time t and $\{2, 0, 1\}$ at time $t + 1$, with every asset in the exact same group both times. A naive label comparison says CMI $= 1.0$, total migration. Hungarian matching recognizes that $0 \mapsto 2$, $1 \mapsto 0$, $2 \mapsto 1$ is just a relabeling and correctly returns CMI $= 0$. This is not a corner case either. Leiden relabels constantly, so without the correction you get systematically inflated migration readings throughout the entire time series.

3.5.2 TDS: Topology Deformation Score

TDS captures structural change at the graph level rather than the cluster-assignment level:

$$\mathrm{TDS}(t) = \alpha \cdot z(W_{\text{degree}}(t)) + \beta \cdot z(J_{\text{community}}(t)) + \gamma \cdot z(S_{\text{spectral}}(t)) \tag{2}$$

The three components are:

- W_{degree}: Wasserstein-1 distance between the degree distributions of consecutive graphs
- $J_{\text{community}}$: community divergence measured as $1 - \mathrm{NMI}(\text{communities}_t, \text{communities}_{\text{baseline}})$
- S_{spectral}: Wasserstein distance between the Laplacian eigenvalue spectra of consecutive graphs

Each component gets passed through a running z-score normalization ($z(\cdot)$) before combination. We had to introduce this because the three components live on completely different natural scales. Degree distance might range from 0 to 5, NMI distance from 0 to 1, and spectral distance from 0 to 0.3. Summing them raw is meaningless since the degree component would dominate everything. The z-scores put them on a common footing.

The spectral component also needed a fix. Our original implementation used zero-padded L_2 norm on the Laplacian eigenvalues, but this destroys spectral gap structure when graph sizes change across windows (which they do, since assets with insufficient history get excluded from early windows). Wasserstein distance handles different-length spectra naturally, and we normalize by the theoretical max eigenvalue of 2.0 for the normalized Laplacian.

3.5.3 AMF: Asset Migration Frequency

While CMI gives a system-wide migration reading per window, AMF tells us which specific assets are the most structurally unstable over time:

$$\mathrm{AMF}_i = \frac{1}{T} \sum_{t=1}^{T} \left[1 - \delta(\sigma_t(c_i(t))\,,\ c_i(t-1))\right] \tag{3}$$

This uses per-step Hungarian matching for consistency with CMI. The results were intuitive: COLO (Colombia, 0.216), DBA (Agriculture, 0.208), and USO (Oil, 0.206) are the most migratory. These are all assets with idiosyncratic drivers (commodity prices, local politics, supply disruptions) that periodically decouple them from their normal cluster. On the other end, VTV (US Value, 0.016) and DIA (Dow, 0.018) barely move at all. They are anchored to the core US equity cluster and almost never leave it.

3.5.4 CPS: Cluster Persistence Score

CPS measures how well a specific cluster holds together between consecutive windows:

$$\mathrm{CPS}_k(t) = \frac{|C_{\sigma(k)}(t) \cap C_k(t-1)|}{|C_{\sigma(k)}(t) \cup C_k(t-1)|} \tag{4}$$

This is a Hungarian-matched Jaccard overlap. High CPS means the cluster retained its members. Low CPS means it fragmented, merged with another cluster, or had significant membership turnover.

3.6 Layer 6: Directional Information Flow

Once you establish that clusters are migrating, the next question is: who moves first? We address this with two complementary approaches.

Net transfer entropy gives us a directed measure of information flow between individual assets:

$$\mathrm{NTE}(X \to Y) = \mathrm{TE}(X \to Y) - \mathrm{TE}(Y \to X) \tag{5}$$

Positive NTE means X is a net sender of information to Y. We compute this for all pairwise combinations using the KSG k-NN estimator, producing a directed flow network that reveals which assets lead and which follow.

The second approach operates at the layer level rather than the asset level. We test whether the CMI time series from the tail-dependence layer Granger-causes the CMI from the Pearson layer. The result was statistically significant at $p = 0.041$, meaning structural changes in the tail-dependence network tend to precede changes in the correlation network. In practical terms: the crash structure reshuffles first, and the normal-conditions correlation structure follows.

The practical takeaway is simple: if you are monitoring tail-dependence CMI and see it spiking while Pearson CMI stays flat, the correlation structure is probably about to shift.

3.7 Layer 7: Statistical Robustness

None of these findings mean much if they do not survive rigorous statistical testing. We built a comprehensive robustness framework covering five distinct validation approaches:

Table 3: Robustness Methods

Method	**Reference**	**Purpose**
Walk-forward validation	–	Out-of-sample replication
Block bootstrap	Politis & Romano (1992)	CIs for TE rankings and Granger F-stat
Phase-randomized surrogates	Theiler et al. (1992)	Null: TE from autocorrelation only
IAAFT surrogates	Schreiber & Schmitz (1996)	Preserves spectrum AND distribution
Bonferroni correction	–	FWER for 9,120 pairwise tests
Benjamini-Hochberg	BH (1995)	FDR control
Storey q-value	Storey (2002)	Adaptive FDR with π_0 estimation
Sensitivity sweeps	–	Window, top-k, resolution, quantile
Monte Carlo power analysis	–	Min sample size for TE significance

The multiple testing correction deserves special attention. With 96 assets in the universe, pairwise Granger and transfer entropy tests produce up to $96 \times 95 = 9{,}120$ hypothesis tests. Without correction, the false positive rate would be enormous. Running Bonferroni (very conservative), BH-FDR (moderate), and Storey q-value (adaptive) side by side lets us see which relationships hold under each level of stringency.

The surrogate testing is also important for the transfer entropy results. Phase-randomized surrogates [5] preserve the power spectrum of the original time series while destroying any nonlinear structure, providing a null hypothesis that says "any apparent TE could be explained by autocorrelation alone." IAAFT surrogates [6] go further by preserving both the spectrum and the marginal distribution.

4 Event Studies

We defined 8 geopolitical event windows across the 16-year dataset, each subdivided into pre-event calibration, escalation buildup, acute shock, counter-response, digestion, and nor-

malization phases. The sub-windowing matters because it lets us track how topology evolves through the full lifecycle of a crisis, rather than just capturing the spike.

The events, in chronological order:

1. European Sovereign Debt Crisis and US Downgrade (2011)
2. COVID-19 Market Crash (March 2020)
3. Fed Hawkish Repricing (2022)
4. SVB Banking Stress (March 2023)
5. Iran-Israel, April 2024 (Operation True Promise I)
6. Japan Carry Trade Unwind (August 2024)
7. Iran-Israel, October 2024 (Operation True Promise II)
8. DeepSeek AI Cost Shock (January 2025)

4.1 Iran-Israel Conflict Cycle

This is our primary case study, and it is ideal for our purposes because it spans three distinct escalation events over 14 months, each producing a different market response.

In April 2024 (True Promise I), Iran launched over 170 drones, 30 cruise missiles, and 120 ballistic missiles at Israel on April 13-14, the first direct military exchange between the two countries. What we observed in the topology data was that the restructuring began during the buildup phase from April 1-12, *before* the actual strikes. By the time the missiles flew, the market had already repositioned itself structurally.

October 2024 (True Promise II) saw approximately 180 ballistic missiles launched on October 1. Israel's counter-strike on October 26 deliberately avoided oil infrastructure. Oil crashed 5-6% on that news, not because the strike was bad, but because the market had priced in something much worse and the restraint triggered a relief unwind.

The June 2025 escalation (Operation Epic Fury / Twelve-Day War) produced our highest TDS reading: 0.176, which actually exceeded COVID. And here is what made it so striking: the Pearson correlation layer showed almost nothing. If we had been running a single-layer analysis, we would have missed the entire structural event. The dCor and tail-dependence layers, however, picked up massive and sustained restructuring. This event alone justified the multi-layer design.

4.2 COVID-19

COVID remains the most extreme single-window reading in our dataset: CMI of 1.0. Every single asset changed cluster assignment in a single step. Complete topological dissolution. But the recovery was surprisingly fast. Within a few weeks, recognizable community structure had re-formed, and the new structure was not radically different from pre-COVID.

Compare that to the geopolitical shocks, which tend to produce lower peak CMI values but much longer-lasting asymmetric shifts. The topology does not snap back the way it does after a symmetric shock like a pandemic selloff. The driving forces (sanctions, conflict, supply chain disruption) persist, and so does the structural change.

5 Key Findings

5.1 Pipeline Results

Before getting into the qualitative findings, here are the top-level numbers from the full pipeline run:

Table 4: Full Pipeline Results (v0.5.0, January 2010 to March 2026)

Metric	**Value**
Assets surviving cleaning	59 of 96
Trading days	3,938
Rolling windows	764 (120-day, 5-day step)
Active clusters (latest window)	6
HMM regimes	3: calm (89.7%), transition (5.1%), stress (5.2%)
Current regime (2026-03-20)	**STRESS**
Mean CMI	0.094
Max CMI	0.441
Mean TDS	0.022
Max TDS	10.195

5.2 Pre-Event Restructuring

The finding we kept seeing across events, and the one we consider most significant, is that markets restructure before the geopolitical event, not during it. Peak topology deformation during the Iran-Israel cycle happened in the buildup to Operation Epic Fury, not when strikes were actually underway. The network was already in its stressed configuration by the time the news broke.

Volatility tells you the event happened. That is useful, but it is retrospective. What the topology data shows is that the repositioning was already underway days before the strikes, which is a different kind of information entirely and arguably the more useful one.

5.3 Multi-Layer Necessity

The June 2025 Twelve-Day War was invisible to Pearson correlation. This is the strongest argument for the multi-layer design in our entire dataset. A single-layer Pearson analysis would have produced a qualitatively wrong conclusion about one of the most significant geopolitical escalations in the sample. The dCor layer picked it up. The tail-dependence layer picked it up even more clearly. But Pearson? Nothing. That is not a theoretical concern about methodology. That is a real event in real data where the standard approach

fails completely.

5.4 Cross-Layer Causality

The tail-dependence CMI Granger-causes the Pearson CMI at $p = 0.041$. In plain terms, the crash structure reshuffles first and the normal-conditions correlation catches up later. That is exactly the ordering you would want if you were trying to build an early warning system out of this: monitor the tail-dependence CMI, and if it starts spiking while Pearson CMI is still flat, correlation structure is probably about to shift.

5.5 Regime-Dependent Leadership

Transfer entropy analysis revealed that the direction of information flow between assets flips depending on the market regime. During calm periods, credit instruments (HYG in particular) and real estate assets tend to lead information flow. During war and stress regimes, the Treasury complex (TLT, GOVT, TIP) takes over as the primary sender.

Looking at the aggregate numbers, HYG (high yield corporate bonds) is the top net information sender overall with a net flow score of +0.385. FXE (euro currency) is the top net receiver. Credit spreads have always been a leading indicator of risk appetite in practice, so seeing HYG dominate net TE is consistent with that. FXE sitting on the receiving end makes sense too, since the euro tends to absorb global risk spillover from pretty much everywhere.

5.6 Migration Characteristics by Asset Class

Table 5: Asset Migration Frequency Extremes

Most Migratory	**AMF**	**Most Stable**	**AMF**
COLO (Colombia)	0.216	VTV (Value)	0.016
DBA (Agriculture)	0.208	DIA (Dow)	0.018
USO (Oil)	0.206	RSPN (Equal Wt Ind.)	0.022

Commodity and frontier emerging market assets consistently show the highest migration frequencies. They sit at the edges of the cluster structure and periodically get pulled into different groups depending on the macro regime. Core US large-cap equity almost never changes cluster assignment. It is structurally anchored.

Fixed income plays a different role entirely. EMB (emerging market bonds) and SHY (short-term Treasuries) have the highest betweenness centrality in the network, meaning they sit on the shortest paths between other clusters more than anything else. They are not the ones migrating, but they are the connective tissue holding the structure together. We did not expect SHY to rank that high, honestly. You would think short-duration Treasuries would be too boring to matter in a network sense, but their role as a parking asset during flight-to-quality flows puts them at the crossroads of everything.

EEM (MSCI Emerging Markets) comes out as the most central asset overall across degree, eigenvector, and closeness centrality. Not surprising, given that EM equity touches essentially everything: commodities through resource exporters, currencies through EM FX, US equities through risk appetite, fixed income through EM spreads. It ends up acting as a hub that amplifies shocks no matter which direction they come from.

6 Methodological Corrections

We want to be upfront about the issues we discovered and fixed during development. Some of these are common pitfalls in the financial networks literature, and documenting them felt important.

CMI permutation invariance. The biggest one. Without Hungarian matching, every time Leiden relabeled clusters (which it does constantly since labels are arbitrary), the CMI metric counted that as real migration. This inflates CMI systematically throughout the entire time series. Our fix uses `scipy.optimize.linear_sum_assignment` to find the optimal label bijection before computing migration.

TDS component incommensurability. The three TDS components (Wasserstein degree distance, NMI community divergence, spectral distance) have completely different natural scales. Combining them without normalization is like adding meters to kilograms. The `TDSNormalizer` class maintains running z-scores for each component so the weighted sum is actually meaningful.

Spectral distance on varying graph sizes. Zero-padded L_2 norm on Laplacian eigenvalues breaks when graphs have different numbers of nodes across windows. Since our asset universe expands as newer ETFs accumulate sufficient history, this happened regularly. Wasserstein distance between eigenvalue distributions handles varying lengths naturally.

MST edge weight double-counting. Summing upper and lower triangles of a symmetric similarity matrix doubles every weight. Small bug, real impact on tree structure. Fixed to take max of symmetric entries.

Granger causality lag selection. Testing multiple lag specifications and selecting the minimum p-value is textbook p-hacking. We apply Bonferroni correction across all tested lags before selecting the best one.

Granger on non-stationary series. Running Granger causality on non-stationary time series produces spurious results. We added an ADF (Augmented Dickey-Fuller) stationarity pre-check; any pair that fails stationarity gets flagged with $p = 1.0$ rather than producing a misleading significant result.

PELT change-point over-detection. The original penalty formula ($\log(n) \cdot \mathrm{Var}(x)$) was too lenient and flagged too many changepoints. Switching to proper BIC ($d \cdot \log(n)$ with $d = 2$) resolved this.

7 Architecture and Pipeline

The full analysis runs as an 8-step CLI pipeline built with Typer for command dispatch and Rich for logging:

1. `fetch-data`: pulls all universe tickers from FMP via async API calls
2. `validate-data`: checks for staleness, coverage gaps, and missing data
3. `build-features`: cleans and aligns prices, computes log returns, saves to parquet
4. `run-clustering`: runs rolling-window Leiden across all 764 windows
5. `run-regimes`: fits the 3-state Gaussian HMM on the topology feature vector
6. `run-migration`: computes CMI, AMF, CPS, and TDS with full graph history
7. `compute-centrality`: betweenness, eigenvector, degree, and closeness per window
8. `export-topology`: writes 4 parquet files for downstream consumption

The codebase totals roughly 6,200 lines of Python spread across 10 source directories, with 32 unit tests covering all core modules.

7.1 Topology Export

The pipeline exports four parquet files designed for reproducibility and integration with downstream ML work:

- `cluster_membership.parquet`: 45,076 rows covering date, ticker, cluster ID, days since last migration, and cluster size
- `centrality_metrics.parquet`: 45,076 rows with degree, betweenness, eigenvector, and closeness centrality per asset per window
- `regime_states.parquet`: 3,918 rows with regime label, probability, and persistence count
- `topology_deformation.parquet`: 763 rows with TDS score, TDS z-score, and layer agreement

8 Comparison to Existing Approaches

Table 6: How Our Framework Addresses Limitations in Prior Work

Existing Approach	Limitation	Our Approach
Static correlation clustering	Ignores temporal dynamics	Rolling windows with regime conditioning
Rolling correlation heatmaps	Pairwise only, no community structure	Community detection with lifecycle tracking
Single-layer networks	Blind to nonlinear and tail effects	Multi-layer graphs with cross-layer agreement
HMM on asset returns	Operates at asset level, not topology	HMM trained on topology state vector
Naive cluster comparison	Label permutations inflate migration	Hungarian matching for true structural change
Raw TDS aggregation	Components on different scales	Running z-score normalization
Standard Granger testing	Lag cherry-picking, no stationarity check	Bonferroni across lags plus ADF pre-check
Single hypothesis testing	Inflated false discovery rate	Bonferroni, BH-FDR, Storey across 9,120 pairs

9 Roadmap and Future Work

The framework as it stands is complete through Phase 5. There are several directions we want to take it.

9.1 Real-Time Extension (Phase 6)

The most natural next step is moving from batch analysis to real-time monitoring. We want a streaming data pipeline that updates rolling windows incrementally, a live dashboard (probably Streamlit or Dash) that tracks CMI, TDS, and layer agreement as they evolve, and an alert system for when tail CMI spikes or TE leadership patterns flip. The pipeline architecture already supports most of this. The streaming ingestion is the straightforward part. The harder problem is computing rolling dependency matrices incrementally without reprocessing the entire window from scratch every time new data comes in.

9.2 Extended Research (Phase 7)

Several extensions would strengthen the research:

- Extending the dataset back to the 2008 Global Financial Crisis using ETFs available from that era
- Incorporating intraday 5-minute returns during crisis windows to see within-day topology evolution
- Applying more rigorous causal discovery methods like PCMCI+ or DYNOTEARS
- Adding a geopolitical NLP layer that ingests GDELT event data or news embeddings as an exogenous signal
- A deeper analysis of crypto assets (BTC, ETH, SOL, DeFi indices) as they become a larger part of the cross-asset landscape

9.3 Robustness Completion

The Phase 4 statistical robustness framework is fully implemented and unit-tested, but running it end-to-end on the complete 96-ETF dataset is still pending. The computational cost is significant: 1,000 bootstrap resamples across 764 windows, multiplied by the pairwise test counts. This will likely need dedicated batch compute time rather than running on a local machine.

10 Conclusion

When we started this project, we assumed the main value would be in the multi-layer design itself. And it was. But the finding we did not expect was how consistently tail dependence leads linear correlation across completely different types of events, from a pandemic selloff

to a Middle Eastern escalation to an AI cost shock. That ordering showed up everywhere, and it ended up being the result with the most practical relevance.

The methodological corrections also turned out to be more important than we initially thought. Hungarian matching for CMI sounds like a minor implementation detail until you realize that without it, your migration readings are systematically inflated across the entire time series. Same story with TDS normalization. These are not edge cases. They change the quantitative results on real data, and the fact that they are rarely addressed in the existing financial networks literature suggests that a lot of published migration analysis may be noisier than it needs to be.

If there is one takeaway from 16 years of data and 8 geopolitical events, it is that watching correlation alone means you are always a step behind. The topology moves first. The tails tell you it is coming. By the time Pearson catches up, the structural shift already happened.

The codebase is open-source and reproducible from a single CLI command. The topology export is designed to plug directly into downstream ML work, which is where we plan to take this next.

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